

# Hybrid Centrality Measures for Service Coverage Problem

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**Abstract.** Service Coverage Problem aims to find an ideal node for installing a service station in a given network such that services requested from various nodes are satisfied while minimizing the response time. Centrality Measures have been proved to be a salient computational science tool to find important nodes in networks. With increasing complexity and vividness in the network analysis problems, there is a need to modify the existing traditional centrality measures. In this paper we propose a new way of hybridizing centrality measures based on node-weighted centrality measures to address the service coverage problem.

**Keywords:** Complex network analysis · Centrality measures · Weighted networks · Hybrid centrality

## 1 Introduction

Flow networks are those networks where information, people, commodities, etc. move from one node to another while traversing on the links. Each node starts with a limited bandwidth of resources for convening the flow and these resources prone to collapse/degrade/reduce over time due to the continuous participation of the node in fostering the flow. Due to this, such networks require uninterrupted maintenance/replenishment service on a regular basis for the proper functioning of the network. Keeping this in mind, service facilities are installed at nodes that meet the service demand of other nodes from time to time.

After a service request is made by some node for more resources or reviving the node after a failure occurs, the service team is sent from the service center to meet the demand. The *response time* to meet the service demand is defined as the time taken between the request for service and start of the service work on the demanding node. The response-time depends on the distance between the node requesting a service and the node with the service stations. It is under the assumption that the service-centers have sufficient resources to meet the demand of other nodes. The *response-efficiency* of a node is inversely correlated

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to the response-time, i.e., when the response-time is least, the node is said to be maximum response-efficient and most suitable for installing service stations. A node with a higher response-time possesses a smaller response-efficiency and is not an appropriate choice for installing service stations.

Given an unweighted network, the objective of *Service Coverage Problem (SCP)* is to find a best-suited node to install a service station facility such that the expected response-time is reduced. In blue other words, the goal is to find a node with the highest expected response-efficiency, i.e. the service should reach faster at the nodes that request for service more often than the nodes with moderate or seldom demands. Therefore, the service center should be installed closer to the nodes with frequent demands.

Randomly choosing a node to install a service facility is certainly not the best solution. It is because the randomly picked node may be very far from the nodes with higher demand. In such a case where a node with a higher load fails, severe damage would have already been caused before the maintenance service reaches it. Thus, we require a measure to evaluate the importance of nodes for the candidacy to install a station covering service requests.

Centrality measures are an important tool in social and complex network analysis to quantify the eminence of nodes. These measures remain invariant under the isomorphic transformation of a network [10]. By definition, a centrality measure is a quantification of the structural importance of a node based on its location, connectivity, or any other structural property. These have been used not only by the network scientists but also by biologists, sociologists, physicists, psychologists, and economists over time. Several measures are coined in literature. The most popular centrality measures for network analysis (traditional measures) are degree, closeness, betweenness and eigenvector centrality. Readers are referred to the books by Jackson [16] and Brandes and Erlebach [10] for a detailed survey on the centrality indices and their applications. There exist several other measures that either extend or generalize these traditional measures or limit them to a restricted application. Moreover, various variants of these centrality measures have been proposed which consider a set of nodes and compute its collective centrality, called *group-centrality* [14]. Yet another direction is to combine various centrality measures to achieve better results for answering more complex problems. Such measures are termed as *Hybrid-centralities* and few of them are summarized in Sect. 2.

In this paper, we propose two hybrid centrality measures to address the service coverage problem. These measures are motivated from the definition of node-weighted centrality measures. Related works to hybrid centrality measures are summarized in Sect. 2. Section 4 introduces the new hybridization of centrality measures for solving the Service Coverage problem based on the definitions of node-weighted centrality measures which are summarized in Sect. 3. The experimental comparison between the newly proposed hybrid centrality measures and traditional centrality measures on real-world networks is comprised in Sect. 5. Finally, in Sect. 6, we conclude and discuss the prospective future directions.

## 2 Related Work

Centrality measures are the tools to find application specific importance of a node. Unweighted Centrality measures mentioned earlier, are the most widely used measures but for complex problems and applications, these measures are inefficient. In that case, a combination of these centrality measures produces better analysis than using them individually. In a recent study [3], authors have proposed a new set of hybrid centralities; Degree-Degree, Degree Closeness, and Degree Betweenness. They noticed that the newly defined centrality measures are different than the traditional centrality measures. On real-world co-authorship networks, they found that all the newly defined hybrid centrality measures are significantly correlated to the authors' performance (sum of citations of authors' h-index). A similar study [2] on weighted collaboration networks was done and three new sets of collaborative centrality measures were proposed. The traditional collaborative centrality measures for an author node (h-index, a-index, g-index) are used to propose new centrality measures. Newly defined set of measures were highly correlated with the traditional performance measures of scholars (Publication Count, Citation Count, h-index, a-index, g-index). Zhang et al. [33] proposed a new hybrid centrality measure to measure a node's importance in satellite communication networks. This measure is also based on the combination of closeness and betweenness centrality but the considered measure in their paper punishes the betweenness importance with a different factor. Qiao et al. [24] proposed a hybrid page scoring algorithm based on degree centrality, betweenness centrality, closeness centrality and the PageRank algorithm. Lee and Djauhari [19] proposed a hybrid centrality measure which was a linear combination of the degree, closeness, betweenness, and eigenvector centrality measures. The proposed measures were used to find the most significant or influential stocks. A hybrid centrality based on the linear combination of degree centrality and cohesion centrality was proposed was Qiu et al. [25] and further used for community detection by Li-Qing et al. [21]. Wang et al. [30] proposed a hybrid centrality measure based on the combination of degree, the degree of neighbors and betweenness. In another study by Buechel and Buskens [12], authors analyze a hybrid centrality model as a combination of extended closeness (a variation of closeness for disconnected networks), betweenness and degree measures. None of the above studies attempt to solve the service coverage problem.

## 3 Node-Weighted Centrality

Most of the centrality measures proposed in literature were first defined for unweighted graphs i.e. all of the nodes and all the edges were assumed homogeneous in the beginning of centrality computation. We refer to these measures as *unweighted centrality measures*. Freeman's [15] Degree, Closeness and Betweenness Centralities and Bonach-ich's [8] Eigenvector Centrality are some popular examples. After realizing the existence and understanding the necessity and importance of weights on the edges, centrality measures for unweighted graphs

were extended to *edge-weighted centrality measures*. These measures take weights on the edges into consideration for ranking the nodes while analyzing the networks but still assuming equal weights on the nodes. A substantial part of the present-day research in the analysis of weighted networks considers only edge weights to determine the topological significance of nodes [22].

One basic reason to ignore the weights on nodes is the hardness and complexity in figuring out the non-trivial mapping from attributes of nodes to real values. Another reason is to avoid the complexity of the analysis process, but sometimes the ignorance of node weights might lead to a wrong analysis of the given network. Therefore, it becomes essential to also consider weights on nodes in the process for a better analysis fully weighted networks. Due to this reason, there was a need to upgrade the previously defined unweighted and edge-weighted centrality measures so that weights at the nodes can also be taken into account for the analysis.

Therefore, while considering the weights on the nodes, there can be two possible extensions of the unweighted and edge-weighted centrality measures:

- *Node-weighted centrality measures*: Consider only the node weights for the analysis while taking all edge weights as one.
- *Fully weighted centrality measures*: Consider both types of weights, edge weights and node weights for the analysis.

In most of the studies done so far on fully weighted networks, while considering weighted edges, weights on the nodes were completely ignored. Meanwhile, only a little work has been accomplished while considering the weights on nodes in the real-world networks [2,3,5,32].

This section summarizes centrality measures that take into consideration the weights on the nodes while giving equal priority to the edges in a given network. These fall in the category of node-weighted centrality measures. We only mention these here to avoid the complexity of including weights on both edges and nodes. Once, these measures are understood, we recommend the readers to combine these measures with the edge-weighted centrality measures given in [22] to derive the definition for fully weighted centrality measures. Let $G = \{V, E\}$ denote an undirected unweighted graph. We add an extra element, weights on the nodes defined as a function $W : V \rightarrow R$, where $R$ is set of real numbers. Let $W_x$ be the weight given at node $x$. We do not directly use $W_x$ in the definitions, but at place of it, we use a function $f$ of $W_x$ (or a function of $W_x$ and $W_y$ depending on the number of parameters) without losing the generality. This gives us the flexibility to tune the function of weights on the nodes according to our need. Here, in this paper we take $f(W_x) = W_x$ for the simplicity. To normalize the new centrality measures, we divide by the maximum possible value that any node can score in the numerator of below-given formulas. We start with degree centrality.

- **Node Weighted Degree centrality**: In [3], weights on nodes are considered and the definition of degree centrality is modified to accommodate the node weights. Abbasi and Hossain [3] considered centrality scores as weights on the nodes. Following it, node weighted centrality of a node $u$ is calculated as:

$\frac{\sum_{v \in V \setminus \{u\}} (f(W_v) \cdot a_{u,v})}{\sum_{x \in V} f(W_x)}$. This measure assigns higher importance to those nodes which are in the immediate neighborhood of highly weighted nodes. Next two measures extend the consideration to all the nodes to compute the eminence of a node.

- **Node Weighted Closeness/Harmonic Centrality**: To target the wider applicability, we define node weighted harmonic centrality (which also can be used in the case of closeness centrality computation as both are highly correlated). Node weighted harmonic centrality of a node $u$ in a network is defined as: $\frac{f(W_u) + \sum_{v \in V \setminus \{u\}} \frac{f(W_v)}{d_{u,v}+1}}{\sum_{x \in V} f(W_x)}$. This measure depends on two factors: weight of the node $u$ under consideration and the effective weights of other nodes corresponding to their distances from node u. It assigns a higher value to the nodes that are of high weights and closer to the nodes with high weights. We refer this measure as *harmonically attenuated node weighted centrality measure.*
- **Node Weighed Decay**: Weighted decay of a node $u$ in a network as defined as $\frac{f(W_u) + \sum_{v \in V \setminus \{u\}} (\delta^{d_{u,v}} \cdot f(W_v))}{\sum_{x \in V} f(W_x)}$ where $\delta$ lies between 0 and 1. Here also the computation of importance depends on the same two factors as in NWCC. But, the contribution of weights of other nodes decays exponentially with distance. Weighted decay assigns a higher value to the nodes that are of high weights and very close to the nodes with high weights. We refer this measure as *exponentially attenuated node weighted centrality measure.*
- **Node Weighted Betweenness Centrality**: In [23], a factor denoting the importance of communication between two pairs is multiplied in the core formula while computing the betweenness centrality. The node weighted betweenness centrality of a node $u$ was defined as $\frac{\sum_{s,t,u \in V : |\{s,t,v\}|=3} f(W_s, W_t) \cdot \frac{\sigma_{st}(u)}{\sigma_{st}}}{\sum_{x,y,u \in V : |\{x,y,v\}|=3} f(W_x, W_y)}$ where $f(W_x, W_y)$ can be assumed to map the weights given on node x and y to a real value denoting the importance of flow happening between x and y.

Eigenvector centrality is a measure where it is still open how to include the effect of node weights.

## 4 New Hybrid Centrality Measures

This section defines two new hybrid measures to solve the Service Coverage Problem. We define these measures for networks without weights on edges. These can be easily further extended to edge-weighted version of hybrid centralities following the ideas for edge-weighted centrality measures given in [22].

Betweenness centrality is used to predict the importance of nodes in a number of flow-based networks e.g. Power Grid networks, Public-transit networks,

Gas Line networks, and Communication networks. The betweenness scores of the nodes in such networks have been considered as the load on the nodes in literature. Several models for understanding and replicating the cascading phenomena have been proposed [13,17]. Some of these models observe that failure of a node with a high load may cause a severe cascading failure and hence result in the breakdown of the whole network system. After a node fails, the best way to reduce the damage is to recover the failed node as soon as possible. We cannot prevent a node from failure but we can definitely put a maintenance mechanism on some of the nodes in the network.

### 4.1 Formulation

In the Service Coverage Problem in flow networks, the load of the nodes (when no other information is provided) can be assumed to be proportional to the betweenness centrality of the nodes. This is because a node with large flow through it, is expected to degrade faster than other nodes and if such a node shuts down, a large amount of flow will be affected. Thus we take the probability of node $j$ requesting for a maintenance service over a fixed time-interval as $Pr[Node\ j\ request\ for\ service] = \frac{BC(j)}{\sum_{k \in V} BC(k)}$, where $BC(j)$ is the betweenness centrality of node $j$. Let $X_{ij}$, be the response-efficiency of a node $i$ to meet service demand from node $j$ if the service station is going to be installed on node $i$. Certainly, $X_{ij}$ depends on the distance between the node requesting for a service (node $j$) and the node with the service stations (node $i$). Harmonic decay can be considered in applications where the nodes are more robust and the service requirement is not urgent while exponential decay might be a more appropriate simplified model in applications like real-time systems where service requests need to be met on an urgent basis.

If the response-efficiency decays harmonically in a given application, i.e. the node is maximum efficient (efficiency value = 1) for itself while it is half efficient (efficiency value = 0.5) for each of its neighbors and so on, then we can formulate for the value of response-efficiency $X_{ij} = \frac{1}{d_{i,j}+1}$. Recall, $d_{u,v}$ denotes the distance, i.e., length of the shortest path from node $u$ to node $v$ in the given network. Let $\chi_i$ be the total response-efficiency of a node $i$ if the service station is going to be installed on node $i$. Then, $\chi_i = \sum_{j \in V} X_{ij} \cdot Pr[Node\ j\ request\ for\ service]$. The expected response-efficiency of node $i$ ($E[\chi_i]$) is computed by taking expectation over that node's response-efficiency to service all the nodes in network. $E[\chi_i] = \sum_{j \in V} \frac{1}{d_{i,j}+1} \cdot \frac{BC(j)}{\sum_{k \in V} BC(k)}$. We define the following hybrid centrality measures based on the above formulation:

***Harmonically-Attenuated-Betweenness Centrality (HABC)*:** This measure is based on the harmonic attenuation of importance with respect to the distance. The harmonically-attenuated-betweenness centrality of a node $u$ is defined as,

$$HABC(u) = \frac{BC(u) + \sum_{v \in V \setminus \{u\}} \frac{BC(v)}{d_{u,v}+1}}{\sum_{w \in V} BC(w)}$$

where $BC(u)$ is the unweighted betweenness centrality of node $u$. This hybrid measure assigns a value that is calculated based on an averaging function of distances and betweenness centrality. This measure assigns a higher value to the nodes that are of high betweenness and closer to other high betweenness central nodes. This measure solves the SCP problem in flow networks where the response-efficiency decays harmonically.

In few complex systems, response-time plays very crucial roles in the functionality of networks and the response-efficiency decays exponentially. In such networks, the response-efficiency decays faster, i.e., the node is maximum efficient (efficiency value = 1) for itself while it decays by a multiplicative factor $\alpha$ (efficiency value = $\alpha$) for each of it's neighbors and so on. Here, $\alpha$ is a severity factor that lies in interval [0, 1]. Then we can formulate for the value of response-efficiency $X_{ij} = \alpha^{d_{i,j}}$. Similar to the previous analysis, the expected efficiency of a node $i$ to be a service station, when efficiency decays exponentially by a factor $\alpha$ is $E[\chi_i] = \sum_{j \in V} \alpha^{d_{i,j}} \cdot \frac{BC(j)}{\sum_{k \in V} BC(k)}$. We define the following hybrid centrality measures based on the above formulation:

***Exponentially-Attenuated-Betweenness Centrality (EABC):*** This measure is based on an exponential attenuation in the power of distance. The Exponentially-attenuated-betweenness centrality of a node $u$ is defined as,

$$EABC(u) = \frac{BC(u) + \sum_{v \in V \setminus \{u\}} \alpha^{d_{u,v}} \cdot BC(v)}{\sum_{w \in V} BC(w)}$$

where $BC(u)$ is the betweenness centrality of node $i$. $\alpha$ is the attenuation factor that is used in the exponential function to the power of distance. It is used to sharply punish the service time. This hybrid measure assigns a value that is calculated based on betweenness centrality of nodes and an exponential function in the power of distances. This measure assigns a higher value to the nodes that are of high betweenness and very close to other high betweenness central nodes. This centrality measure is a sharp attenuation variant of harmonically-attenuated-degree centrality. This measure solves the service coverage problem when the service is required on a very urgent basis.

Let $m$ be the number of links and $n$ be the number of nodes in a given network. The over all time to compute proposed hybrid centrality measure is $O(mn)$ in unweighted and $O(mn + n^2 \log n)$ in weighted graphs. It is due to the time complexity for computing betweenness centrality [9]. Efficient algorithms for updating and estimating betweenness centrality measures in dynamic and large-size networks are discussed in [4,27,28].

## 5 Experimental Results

In this section, we discuss the simulation results on various real-world networks. First, we discuss the experimental setup. Then, we mention all the data set used for experimentation. Next, we provide a comparison of traditional centrality measures: degree (DC), closeness (CC), betweenness (BC) and, eigenvector (EC)

**Table 1.** Considered real-world networks

| Instance name | n | Avg. Deg. | $\hat{C}$ | Network type |
|---|---|---|---|---|
| Chicago Road [18] | 1467 | 1.7696 | 0.0001 | Transport network |
| Chilean Power Grid [29] | 347 | 2.55908 | 0.0865 | Energy network |
| Euroroad [18] | 1174 | 2.41397 | 0.0167 | Transport network |
| London Metro [6] | 266 | 2.31579 | 0.0363 | Transport network |
| Madrid Metro [6] | 209 | 2.29665 | 0.0056 | Transport network |
| Mexico Metro [6] | 147 | 2.23129 | 0.0034 | Transport network |
| Minnesota Road [26] | 2642 | 2.50038 | 0.016 | Transport network |
| Moscow Metro [6] | 134 | 2.32836 | 0.0174 | Transport network |
| New York Metro [6] | 433 | 2.194 | 0.0173 | Transport network |
| Oldenburg Road [11] | 6105 | 2.3027 | 0.0108 | Transport network |
| Openflights [6] | 2939 | 10.66825 | 0.4526 | Transport network |
| Osaka Metro [6] | 108 | 2.27778 | 0.0001 | Transport network |
| p2p-Gnutella08 [20] | 6301 | 6.59483 | 0.0109 | Internet peer-to-peer network |
| Paris Metro [6] | 299 | 2.38127 | 0.0204 | Transport network |
| as20000102 [20] | 6474 | 4.29255 | 0.2522 | Autonomous systems graph |
| Seoul Metro [6] | 392 | 2.22959 | 0.006 | Transport network |
| Shanghai Metro [6] | 148 | 2.13514 | 0.0029 | Transport network |
| Tokyo Metro [6] | 217 | 2.41475 | 0.0237 | Transport network |
| US Air [7] | 332 | 12.80723 | 0.6252 | Transport network |
| US Power Grid [31] | 4941 | 2.6691 | 0.0801 | Energy network |
| NRPG Data [1] | 246 | 3.03252 | 0.1071 | Energy network |

and the proposed hybrid centrality measure (HABC, EABC($\alpha = 0.5$), EABC($\alpha = 0.75$)) in the considered networks. The experiments are performed on a Linux system with 8 GB RAM and Intel Core i5-8250U CPU at 1.6 GHz. Implementation is done in Python 3.6.7 with the help of Networkx library.

**Real-World Data Sets:** The proposed solution to solve service coverage problem discussed in this paper hybridizes betweenness centrality within closeness/harmonic and decay centrality. Betweenness centrality is first used to compute the load on each node, therefore, we have selected 26 real-world flow networks. We provide a brief summary of these networks in Table 1 and [1,6,7,11,18,20,26,29,31] can be referred for a detailed description of the networks. We have considered various types of transport networks, energy networks, internet peer to peer network, etc. The columns of Table 1 consist of names of the network instances, the number of nodes ($n$), the average degree of the networks (Avg. Deg.), average clustering coefficient ($\hat{C}$) and the network type respectively.

**Simulation:** We have conducted simulations for evaluating the performance of various traditional centrality measures and the hybrid measures proposed in this

**Table 2.** Expected response-time of the proposed hybrid measures and the traditional centrality measures when the most central node is made the service center

| Instance name | BC | DC | CC | EC | HABC | EABC($\alpha = 0.5$) | EABC($\alpha = 0.75$) |
|---|---|---|---|---|---|---|---|
| Chicago Metro | **0.49** | **0.49** | **0.49** | **0.49** | **0.49** | **0.49** | **0.49** |
| Chilean Power Grid | **0.29** | 0.33 | **0.29** | 0.33 | **0.29** | **0.29** | **0.29** |
| Euroroad | **0.88** | **0.88** | 0.89 | 1.05 | 0.88 | **0.88** | **0.88** |
| London Metro | 0.83 | 0.83 | **0.80** | 0.83 | **0.80** | **0.80** | **0.80** |
| Madrid Metro | 0.46 | 0.46 | 0.44 | 0.66 | 0.46 | 0.46 | **0.42** |
| Mexico Metro | **0.50** | **0.50** | 0.54 | **0.50** | **0.50** | **0.50** | 0.54 |
| Minnesota Road | 2.29 | 3.21 | 2.15 | 3.11 | 3.29 | **2.29** | 3.45 |
| Moscow Metro | 0.40 | 0.40 | **0.37** | 0.39 | 0.39 | 0.39 | **0.37** |
| Newyork Metro | **1.11** | 1.3 | 1.12 | 1.3 | **1.11** | **1.11** | **1.11** |
| Oldenburg Road | **1.66** | **1.66** | 1.68 | 2.64 | **1.66** | **1.66** | **1.66** |
| Openflights | **0.16** | **0.16** | **0.16** | **0.16** | **0.16** | **0.16** | **0.16** |
| Osaka Metro | 0.48 | 0.48 | 0.44 | 0.48 | 0.48 | 0.48 | **0.42** |
| p2p-Gnutella08 | **0.29** | **0.27** | **0.27** | **0.27** | **0.27** | **0.27** | **0.27** |
| Paris Metro | 0.55 | 0.55 | 0.55 | 0.56 | **0.51** | **0.51** | **0.51** |
| as20000102 | **0.12** | **0.12** | **0.12** | **0.12** | **0.12** | **0.12** | **0.12** |
| Seoul Metro | 1.06 | 1.25 | 1.06 | 1.23 | **1.02** | **1.02** | **1.02** |
| Shanghai Metro | **0.53** | **0.53** | 0.64 | **0.53** | 0.64 | **0.53** | 0.64 |
| Tokyo Metro | 0.57 | 0.57 | 0.57 | 0.57 | 0.57 | 0.57 | **0.49** |
| US Air | **0.10** | **0.10** | **0.10** | **0.10** | **0.10** | **0.10** | **0.10** |
| US Power Grid | 0.92 | 1.34 | **0.83** | 2.12 | **0.83** | **0.83** | **0.83** |
| NRPG | **0.30** | 0.45 | **0.30** | 0.37 | **0.30** | **0.30** | **0.30** |

paper which have been summarized in Table 2. The performance of a centrality measure is evaluated by computing the expected response-time in terms of the average distance of the service requesting nodes from the top central node as per this centrality measure.

Betweenness centrality has been used as one of the best measure to map loads on nodes in flow networks [17]. A node having higher load will be more frequent in requesting for services. Therefore, we consider the probability of a node requesting for services proportional to the betweenness centrality of that node. The expected response-time is computed over $\lceil n/10 \rceil$ service requests where $n$ is the number of nodes in respective real-world networks. The first column in the table contains label of the considered real-world network instances. The next columns lists the expected response-time of the traditional centrality measures (BC, CC, DC, EC) and the proposed hybrid measures (HABC, EABC($\alpha = 0.5$), and EABC($\alpha = 0.75$)).

It is evident from Table 2 that no traditional measure can consistently find an ideal service center. While at least one of the proposed hybrid centrality measures are always able to minimize the average response-time for all considered networks. In some cases such as Madrid Metro, Paris Metro, Osaka Metro,

**Table 3.** Ranking list of top three central nodes using the proposed hybrid measures and the traditional centrality measures on considered real-world networks given in Table 1

| Instance Name | HABC | | | EABC($\alpha = 0.5$) | | | EABC($\alpha = 0.75$) | | | BC | | | CC | | | DC | | | EC | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | R1 | R2 | R3 | R1 | R2 | R3 | R1 | R2 | R3 | R1 | R2 | R3 | R1 | R2 | R3 | R1 | R2 | R3 | R1 | R2 | R3 |
| **Chicago Road** | 1156 | 1157 | 1147 | 1156 | 1157 | 1147 | 1156 | 1157 | 1147 | 1156 | 1157 | 1147 | 1156 | 1157 | 1147 | 552 | 922 | 10 | 1156 | 1147 | 1150 |
| **Chilean Power Grid** | 178 | 105 | 147 | 178 | 105 | 147 | 178 | 105 | 147 | 178 | 105 | 108 | 178 | 105 | 147 | 108 | 6 | 178 | 108 | 264 | 29 |
| **Euroroad** | ***402*** | 284 | 277 | 402 | 284 | 277 | 402 | 284 | 277 | ***402*** | 284 | 277 | 401 | 402 | 403 | 284 | 7 | 39 | 7 | 43 | 454 |
| **London Metro** | ***115*** | 29 | 12 | 115 | 29 | 12 | 115 | 29 | 12 | 12 | 115 | 180 | ***115*** | 29 | 214 | 12 | 164 | 305 | 214 | 115 | 219 |
| **Madrid Metro** | ***80*** | 162 | 146 | 80 | 162 | 146 | 162 | 80 | 146 | ***80*** | 146 | 162 | 162 | 80 | 235 | 80 | 15 | 9 | 15 | 54 | 40 |
| **Mexico Metro** | ***26*** | 70 | 23 | 26 | 70 | 78 | 70 | 26 | 54 | ***26*** | 70 | 25 | 70 | 26 | 23 | 26 | 127 | 61 | 26 | 70 | 23 |
| **Minnesota Road** | 300 | 328 | 280 | 1821 | 2069 | 300 | 280 | 328 | 300 | 1821 | 2069 | 2063 | 1356 | 1820 | 1109 | 2418 | 35 | 32 | 1927 | 1930 | 1987 |
| **Moscow Metro** | 59 | 71 | 25 | 59 | 71 | 25 | 71 | 59 | 73 | 25 | 74 | 59 | 73 | 71 | 25 | 24 | 59 | 21 | 59 | 25 | 73 |
| **New York Metro** | ***424*** | 191 | 410 | 424 | 191 | 410 | 424 | 191 | 410 | ***424*** | 191 | 148 | 84 | 424 | 410 | 273 | 228 | 410 | 273 | 228 | 188 |
| **Oldenburg Road** | ***831*** | 714 | 731 | 831 | 714 | 711 | 831 | 714 | 803 | ***831*** | 714 | 593 | 714 | 731 | 712 | 831 | 1657 | 1571 | 1677 | 1683 | 1691 |
| **Openflights** | 53 | 65 | 57 | 53 | 57 | 65 | 53 | 57 | 65 | 53 | 65 | 1576 | 53 | 65 | 57 | 53 | 65 | 59 | 53 | 59 | 65 |
| **Osaka Metro** | ***25*** | 70 | 95 | 25 | 70 | 95 | 70 | 95 | 25 | ***25*** | 99 | 70 | 95 | 70 | 25 | 25 | 54 | 42 | 25 | 81 | 54 |
| **p2p-Gnutella08** | ***123*** | 367 | 127 | 123 | 367 | 127 | 123 | 127 | 367 | 5831 | 1317 | 424 | ***123*** | 127 | 1317 | 123 | 127 | 367 | 367 | 123 | 145 |
| **Paris Metro** | 150 | 57 | 235 | 150 | 57 | 235 | 150 | 57 | 235 | 57 | 150 | 106 | 57 | 150 | 235 | 57 | 177 | 246 | 186 | 150 | 235 |
| **as20000102** | 2 | 10 | 7 | 2 | 10 | 7 | 2 | 10 | 7 | 2 | 7 | 10 | 2 | 10 | 7 | 2 | 10 | 7 | 2 | 10 | 7 |
| **Seoul Metro** | 220 | 480 | 357 | 220 | 444 | 391 | 220 | 480 | 357 | 220 | 444 | 391 | 220 | 480 | 357 | 254 | 444 | 391 | 103 | 254 | 125 |
| **Shanghai Metro** | ***136*** | 16 | 113 | 136 | 16 | 213 | 136 | 113 | 16 | 16 | 136 | 65 | ***136*** | 113 | 97 | 16 | 136 | 220 | 16 | 137 | 138 |
| **Tokyo Metro** | 150 | 85 | 53 | 150 | 85 | 53 | 85 | 150 | 60 | 150 | 53 | 85 | 150 | 60 | 85 | 150 | 53 | 120 | 150 | 120 | 36 |
| **US Air** | 118 | 261 | 201 | 118 | 201 | 47 | 118 | 201 | 47 | 118 | 8 | 261 | 118 | 261 | 67 | 118 | 261 | 255 | 118 | 261 | 255 |
| **US Power Grid** | ***1378*** | 1365 | 2685 | 1378 | 1365 | 2685 | 1378 | 1365 | 1678 | 651 | 559 | 1365 | ***1378*** | 1678 | 2944 | 2847 | 602 | 932 | 4422 | 4436 | 4419 |
| **NRPG Data** | ***233*** | 235 | 238 | 233 | 235 | 238 | 233 | 235 | 238 | 70 | 235 | 213 | ***233*** | 238 | 70 | 194 | 65 | 56 | 239 | 213 | 215 |

Seoul Metro, and Tokyo Metro, the proposed measures even outperform all the traditional measures.

**Node Ranks:** Here, we discuss the ranking result of nodes by various centrality measures. Table 3 contains the ranking list of the top three central nodes using the proposed hybrid measures and the traditional centrality measures on the considered real-world networks given in Table 1. Due to space constraints, only the data of top three central nodes have been presented in the paper. The first column in the table contains the label of the considered real-world network instances. The next columns in a group of three lists top 3 central nodes using the proposed hybrid measures (HABC, EABC($\alpha = 0.5$), and EABC($\alpha = 0.75$)) and the traditional centrality measures (BC, CC, DC, EC).

Finding only the most central node might not be useful in the cases when the top central node does not allow it to be made as a service facility due to some constraints. In that case, finding the first top-$k$ potential centers are important. It is evident from the experimental results that the proposed measures are results of hybridizations between closeness centrality and betweenness centrality.

We have emphasized the top-ranking node for a few entries in the table and have written them in bold to exhibit the above phenomenon. For some networks, the top-ranking node as per HABC is the same as BC but not as CC and vice-versa. The ranking on two networks (Minnesota Road and Moscow Metro) also provide evidence that top-ranking nodes due to the proposed centrality measures are different from closeness and betweenness centrality. In addition to these two, another network (Openflights) shows that these proposed measures also rank nodes differently than each other.

**Table 4.** Spearman's rank correlation between the proposed and the traditional centrality measures and on considered real-world networks given in Table 1

| | HABC | | | | EABC($\alpha = 0.5$) | | | | EABC($\alpha = 0.75$) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Instance Name** | **BC** | **CC** | **DC** | **EC** | **BC** | **CC** | **DC** | **EC** | **BC** | **CC** | **DC** | **EC** |
| **Chicago Road** | 0.2087 | 0.3688 | 0.2393 | 0.2027 | 0.2040 | 0.3437 | 0.2351 | 0.2024 | 0.2035 | 0.3241 | 0.1985 | 0.1515 |
| **Chilean Power Grid** | 0.0691 | -0.0620 | 0.0132 | 0.0057 | 0.0224 | 0.0274 | 0.0506 | 0.0163 | 0.0929 | 0.0867 | -0.0200 | 0.0850 |
| **Euroroad** | 0.7707 | 0.0144 | 0.2019 | -0.0010 | 0.7985 | 0.0190 | 0.1675 | -0.0174 | 0.7656 | 0.0214 | 0.1834 | -0.0227 |
| **London Metro** | 0.1625 | 0.1450 | 0.0296 | 0.0312 | -0.0291 | 0.0323 | 0.0917 | 0.0636 | 0.0432 | 0.1496 | -0.0837 | 0.0324 |
| **Madrid Metro** | 0.0540 | 0.1561 | -0.0912 | 0.0149 | 0.0284 | 0.0891 | -0.0949 | 0.1213 | 0.0109 | 0.0814 | -0.1316 | 0.0049 |
| **Mexico Metro** | -0.0598 | 0.2043 | 0.0428 | 0.1145 | 0.1267 | -0.0479 | 0.0516 | 0.1401 | -0.0037 | 0.2043 | -0.0716 | 0.1522 |
| **Minnesota Road** | -0.0068 | -0.0358 | -0.0777 | -0.1648 | 0.0286 | -0.0221 | 0.0075 | -0.0393 | 0.0015 | -0.0552 | 0.0345 | -0.1444 |
| **Moscow Metro** | 0.0824 | 0.0474 | 0.1245 | 0.0570 | -0.1814 | -0.1215 | -0.2326 | 0.0627 | 0.1012 | 0.0468 | 0.0331 | -0.1045 |
| **New York Metro** | 0.0386 | 0.0865 | -0.0061 | 0.0760 | 0.0599 | 0.0394 | -0.0382 | -0.0222 | 0.0083 | -0.0139 | -0.0651 | -0.0156 |
| **Oldenburg Road** | 0.0098 | 0.1332 | 0.0094 | -0.0222 | 0.0075 | 0.0593 | 0.0202 | 0.0169 | 0.0113 | 0.0991 | 0.0039 | -0.0083 |
| **Openflights** | 0.0018 | 0.1090 | 0.0263 | 0.0799 | 0.0055 | 0.1417 | 0.0318 | 0.0898 | 0.0032 | 0.1305 | 0.0200 | 0.0618 |
| **Osaka Metro** | 0.0110 | 0.1728 | 0.0262 | 0.1440 | 0.0482 | 0.1662 | 0.0402 | 0.1398 | 0.0795 | 0.1031 | -0.0713 | 0.0656 |
| **p2p-Gnutella08** | 0.0537 | 0.0760 | 0.0880 | 0.0491 | 0.0636 | 0.0634 | 0.0775 | 0.0485 | 0.0660 | 0.0529 | 0.0754 | 0.0561 |
| **as20000102** | 0.1327 | 0.5878 | 0.1837 | 0.4513 | 0.1408 | 0.5881 | 0.1810 | 0.4423 | 0.1309 | 0.6273 | 0.1912 | 0.4392 |
| **Seoul Metro** | 0.0520 | -0.0055 | -0.0048 | -0.0389 | 0.0257 | 0.0143 | 0.0480 | 0.0775 | -0.0427 | 0.0109 | -0.0643 | 0.0136 |
| **Shanghai Metro** | 0.1104 | -0.0165 | 0.1168 | 0.0977 | 0.0721 | 0.1124 | 0.0326 | 0.1617 | 0.0014 | 0.0922 | 0.1809 | 0.0610 |
| **US Air** | 0.0855 | 0.1707 | 0.0522 | 0.0608 | 0.1171 | 0.2246 | 0.0059 | 0.0674 | 0.0503 | 0.2372 | 0.0243 | 0.1065 |
| **US Power Grid** | 0.0450 | 0.0831 | 0.0830 | 0.0437 | 0.0350 | 0.0279 | 0.0345 | 0.0038 | 0.0436 | 0.0598 | 0.0622 | 0.0436 |
| **Paris Metro** | -0.1077 | 0.0421 | -0.0625 | 0.1169 | 0.0302 | 0.0846 | -0.0271 | 0.0351 | 0.0175 | 0.1305 | -0.0914 | 0.0309 |
| **Tokyo Metro** | 0.1693 | 0.0920 | 0.0064 | 0.0460 | 0.0555 | 0.0883 | 0.0958 | 0.0511 | 0.0646 | 0.1071 | -0.0794 | 0.1039 |
| **NRPG Data** | 0.0118 | 0.1334 | -0.0693 | 0.0934 | 0.0445 | 0.0975 | -0.1389 | 0.1395 | -0.0867 | 0.1659 | -0.1322 | 0.0936 |
| Average Correlation | 0.0902 | 0.1192 | 0.0444 | 0.0694 | 0.0811 | 0.0966 | 0.0305 | 0.0858 | 0.0744 | 0.1268 | 0.0094 | 0.0574 |
| Standard Deviation | 0.1729 | 0.1441 | 0.0907 | 0.1152 | 0.1802 | 0.1485 | 0.1043 | 0.1042 | 0.1701 | 0.1432 | 0.1069 | 0.1139 |

It is evident that no single standard centrality measure is a perfect substitute for our proposed centrality measures. As our measures follow directly from the requirements of the Service Coverage Problem, it is evident that current centrality measures are not sufficient to solve the problem adequately.

The popular Spearman's rank correlation coefficient is computed to evaluate the correlation between the proposed hybrid centralities and the four traditional centrality measures on the considered real-world networks. These results are summarized in Table 4. The first column in the table contains the label of the considered real-world network instances. The next four columns comprise of the rank correlation between HABC and the traditional centrality measures (BC, CC, DC, EC). Similarly the next four columns contain the rank correlation values between and EABC($\alpha = 0.5$) and BC, CC, DC, EC respectively. The last four columns are the rank correlation between EABC($\alpha = 0.75$) and the traditional centrality measures.

The experimental results in Table 4 make it evident that ranking by the proposed hybrid measures is different than traditional centrality measures. The average correlation coefficient between the proposed measures and traditional measures, although is positive but very small. The standard deviation is larger than the average coefficient values for most of the computation of rank correlation. It is due to several negative correlation coefficient values. Based on the average rank correlation coefficient values, HABC is best correlated with CC and then BC among the traditional measures. EABC($\alpha = 0.75$) also exhibits similar pattern as the decay rate is slower than EABC($\alpha = 0.5$). EABC($\alpha = 0.5$) is best

correlated with the CC and then EC among the traditional measures. The proposed measures are least correlated with DC. Therefore, the existing traditional measures do not provide the solution to the service coverage problem. The top-ranked nodes by the proposed hybrid centrality measures are more appropriate in this application.

## 6 Conclusion

In this paper, we have proposed two new hybrid centrality measures based on closeness (harmonic), betweenness and decay measures. The hybridization is used to solve service coverage problem in flow networks where the demand for services is assumed to be proportional to the betweenness centrality. The proposed hybridization of centrality measures is based on the node weighted centrality measures. The experimental results on several real-world networks show that the proposed measures rank the nodes differently than traditional measures. Our measures perform well in simulations, and give statistically different results than the traditional centrality measures. The proposed measures can also be used in another application that requires installing a facility farthest from the failure prone nodes. The solution, in this case, will be the node with the minimum expected response-efficiency. Analyzing these measures on various other real-world networks and at the place of betweenness, hybridizing other measures specific to some particular applications are the possible future directions. The analysis of real-world networks where non-uniform weights are given at nodes using node weighted centrality measures is another open direction.

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